

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

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ENTERPRISE LOYALTY REWARDS MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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*In partial fulfilment for the award of the degree
of*

**BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING**

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August 2025

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SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals are a collection of 17 global goals designed to be a blueprint to achieve a better and more sustainable future for all. The SDGs, set in 2015 by the United Nations General Assembly and intended to be achieved by the year 2030. In 2015, 195 nations agreed as a blueprint that they can change the world for the better. The project is based on the relevant global goals.

Questions	Answer Samples
Which SDGs does the project directly address?	SDG 8: Decent Work and Economic Growth & SDG 9: Industry, Innovation, and Infrastructure – by promoting digitalized retail transaction processing, transparent rewards management, and secure financial oversight.
What strategies or actions are being implemented to achieve these goals?	Digital enterprise loyalty platform to eliminate paper-based tracking, automate point calculations, and ensure fraud detection across customer transactions.
How is progress measured and reported in relation to the SDGs?	Tracking active user engagement, customer retention rates, point redemption efficiency, and automated audit logs.
How were these goals identified as relevant to the project's objectives?	Helps enterprises manage customer retention, streamline rewards distribution, eliminate fraud, and support sustainable digital business practices.
Are there any partnerships or collaborations in place to enhance this impact?	Working with retail partner merchants, financial analytics providers, and secure payment processing services.

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BONAFIDE CERTIFICATE

Certified that this project report titled “**Enterprise Loyalty Rewards Management System**” is the Bonafide work of **KARTHICK GEETHANATH (727723EUCS501)** who carried out the project work under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

At this juncture, we take the opportunity to convey our sincere thanks and gratitude to the management of the college for providing all the facilities to us.

We wish to convey our gratitude to our college principal, **Dr. K. Porkumaran** for forwarding us to do our project and offering adequate duration to complete our project.

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ABSTRACT

The rapid growth of retail commerce and digital services has increased the demand for efficient, scalable customer loyalty and rewards management platforms. Traditional paper-based or disjointed reward tracking methods often suffer from points accounting inaccuracies, lack of transparency, fraud vulnerability, and difficulties in monitoring customer retention, making it challenging for administrators and staff to manage customer engagement effectively. This project aims to develop a comprehensive, full-stack **Enterprise Loyalty Rewards Management System** built using ReactJS, Java Spring Boot, Spring Security, and PostgreSQL that simplifies and streamlines the entire customer rewards lifecycle.

The system provides a centralized multi-role platform enabling Customers to track points and redeem tier rewards, Staff to allocate purchase points seamlessly, Loyalty Managers to configure reward tiers and challenges, and Super Admins to oversee security audit logs, fraud analytics, and system announcements. Key features include an automated FIFO (First-In, First-Out) points expiry engine, real-time fraud detection rules, dynamic tier calculation (Bronze, Silver, Gold, Platinum), role-based JWT authentication, and administrative CSV/PDF export tools. By providing strict role-based access control (RBAC) and traceable point accounting, the platform eliminates double-crediting, prevents unauthorized account modifications, and delivers an intuitive, reliable experience for all enterprise stakeholders.

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LIST OF ABBREVIATIONS

S. No	ABBREVIATIONS	EXPANSION
1	RBAC	Role-Based Access Control
2	RAM	Random Access Memory
3	GB	Giga Bytes
4	VS Code	Visual Studio Code
5	OS	Operating System
6	HTTP	Hyper Text Transfer Protocol
7	JPA	Java Persistence API
8	API	Application Programming Interface
9	JDBC	Java Database Connectivity
10	SQL	Sequential Query Language
11	UI	User Interface
12	DOM	Document Object Model
13	JSX	JavaScript XML
14	JWT	JSON Web Token
15	UML	Unified Modelling Language
16	DFD	Data Flow Diagram
17	FIFO	First-In, First-Out

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The rapid expansion of retail enterprises and digital service management has increased the demand for efficient and organized customer loyalty management. Traditional rewards methods often lead to point calculation discrepancies, inefficient tier tracking, manual errors, and security vulnerabilities. These challenges not only frustrate customers but also complicate operational governance for enterprise store managers and system administrators.

The Enterprise Loyalty Rewards Management System project aims to address these critical issues by providing a centralized, user-friendly, and multi-role platform for customers, staff, loyalty managers, and super administrators. The system simplifies the process of earning purchase points, redeeming tier-specific rewards, tracking point expiration via FIFO logic, and monitoring transaction history, while enabling administrators to govern role permissions, detect fraud, and broadcast announcements.

This platform allows customers to view real-time point balances, track tier progress (Bronze, Silver, Gold, Platinum), and manage their profiles conveniently. At the same time, store staff can allocate points for purchases, loyalty managers can create campaigns and store promotions, and super administrators can manage security settings and execute administrative CSV/PDF audit exports.

One of the key benefits of this system is its ability to enforce strict Role-Based Access Control (RBAC), provide real-time transaction updates, and maintain mathematical transparency across all point operations. By overcoming the limitations of traditional loyalty systems, the platform ensures security, operational efficiency, and customer satisfaction for modern enterprise environments.

1.2 COMPONENTS OF SYSTEM

System Landing Page

The landing page serves as the entry point for users, providing an overview of the loyalty rewards portal, membership tiers, available catalog rewards, and security features. It allows users to quickly navigate to login/registration forms and explore system capabilities.

System Dashboard

The dashboard is the central hub tailored for each role (Super Admin, Loyalty Manager, Staff, Customer). Customers view their points balance, recent rewards, and tier progress, while administrators monitor total transactions, fraud flags, and active members.

Member Management Roster

This component enables authorized staff and administrators to search and manage member accounts. Strict role restrictions ensure staff view customers only, loyalty managers access staff and customers, while super admins govern all system accounts.

Points Management Portal

Staff and managers can calculate purchase points (1 point per ₹10 spent) and deduct points for redemptions. The backend automatically records transaction history and applies FIFO lot tracking.

Admin Announcements & Notifications

Super Admins can broadcast system announcements to active Loyalty Managers and Staff. Notifications appear in real-time in the top notification bell dropdown with unread indicators and  badges.

User Profile Management

All registered members can manage their personal profiles, update phone details, change passwords securely, and track individual membership tier statuses.

Booking & Transaction Ledger History

Users and administrators access a complete audit log of transactions including bill amounts, points earned/deducted, performed-by user details, and timestamp records, promoting complete operational transparency.

Security and Access Control

Strong security measures, including Spring Security RBAC, JWT token validation, and password encryption, protect sensitive user data. Controllers explicitly enforce 403 Forbidden responses on unauthorized role access.

1.3 ADVANCED TECHNOLOGIES

FIFO Point Expiry Engine

An automated background lot tracking engine consumes earned points on a First-In, First-Out basis, ensuring points expire accurately according to configured policy dates.

Real-Time Fraud Evaluation

Rule-based algorithms evaluate transactions in real-time, detecting high-value point additions, unusual frequency, or self-crediting attempts by store staff.

Administrative Audit Exports

Super Admins can export filtered transaction history logs into formatted CSV files or OpenPDF-generated official administrative reports for compliance auditing.

Role-Based JWT Authorization

Stateless JWT authentication embeds granted authorities into request headers, allowing the Spring Security filter chain to authorize REST endpoints without session overhead.

Mobile-Responsive Platform

A responsive React frontend powered by Vite and CSS modules enables seamless management on desktop workstations, tablets, and mobile devices.

1.4 GLOBAL PERSPECTIVES

Global perspectives on customer loyalty management systems are shaped by digital transformation, evolving consumer expectations, strict data privacy regulations, and the need for secure transaction processing.

- **Shift Toward Digital Loyalty Platforms:** Worldwide, retail enterprises are replacing physical card programs with digital loyalty platforms that provide real-time rewards, instant point crediting, and multi-tier benefits.
- **Integration of Security and Fraud Analytics:** Enterprise loyalty systems globally are integrating AI and rule-based fraud detection to prevent point inflation, insider abuse, and unauthorized account access.
- **Focus on User Experience and Transparency:** Modern systems emphasize transparent mathematical point tracking, instant notification alerts, and intuitive dashboards to foster customer trust and long-term engagement.
- **Strict Data Governance and Compliance:** Enterprise platforms adhere to international security standards through encrypted credential storage, role-based access control (RBAC), and verifiable transaction audit trails.

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

- The current loyalty management approach in many retail stores relies on manual spreadsheets or standalone point-of-sale add-ons to credit points and track redemptions. This often leads to point discrepancies, delayed updates, and limited visibility for customers.
- Existing digital solutions where available often lack fine-grained Role-Based Access Control (RBAC). Staff members or store managers frequently possess unmonitored privileges to edit balances or assign unauthorized admin roles.
- For facility administrators, existing tools provide minimal audit logging, lacking automated fraud detection rules, FIFO point expiry management, and administrative CSV/PDF reporting tools.
- Security and user privacy are often minimal in traditional systems. Plaintext credential handling or basic authentication exposes transaction data to tampering and unauthorized access.

2.1.1 DRAWBACKS

While legacy digital loyalty software improves basic tracking compared to paper stamps, existing solutions face several significant challenges:

Quality of Service

- **Inconsistent Customer Experience:** Delays in crediting points or resolving redemption issues reduce customer trust and retention.
- **Lack of Transparency:** Customers cannot trace exact calculations for earned or expired points.

Limited Functionality

- **Scope of Features:** Lack of automated FIFO point expiry, real-time fraud flags, and admin announcements.
- **Integration Gaps:** Inability to produce official PDF audit reports or export filtered transaction ledgers.

Security and Privacy

- **Privilege Escalation Risks:** Insecure REST APIs allow lower-level staff to modify admin privileges or credit personal accounts.
- **Data Security Vulnerabilities:** Weak authentication mechanisms expose customer data to unauthorized disclosure.

2.2 PROBLEM DEFINITION

1. **Quality of Service Issues:** Legacy loyalty management tools suffer from inconsistent point crediting, poor user interfaces, and delayed transaction updates, resulting in customer dissatisfaction and operational bottlenecks.
2. **Security & Authorization Deficiencies:** Lack of strict backend-enforced RBAC enables unauthorized users to view elevated profiles, alter account roles, or manage points outside assigned privileges.
3. **Manual Errors & Fraud Exposure:** Absence of real-time fraud monitoring and automated point expiry rules allows point duplication, unauthorized staff crediting, and loss of accounting auditability.

2.3 PROPOSED SYSTEM

- **User-Friendly Multi-Role Portal:** Intuitive React frontend tailored specifically for Super Admin, Loyalty Manager, Staff, and Customer roles.
- **Strict Backend Role-Based Access Control:** Spring Security controllers strictly enforce HTTP 403 Forbidden on unauthorized endpoint calls across all 4 roles.
- **Automated FIFO Points & Tier Engine:** Automatic calculation of points (1 pt / ₹10 spent), dynamic tier evaluation (Bronze to Platinum), and background FIFO lot expiry.
- **Real-Time Fraud & Announcement Engine:** Automated fraud flag evaluation for unusual transactions and Super Admin announcement broadcasting to staff/managers with 🔔 notification bell integration.
- **Administrative Audit & Export:** Filtered transaction ledger search with official CSV and OpenPDF export capabilities.

2.3.1 ADVANTAGES

- Complete Mathematical Transparency and Auditability
- Strict Backend-Enforced Security & RBAC Guardrails
- Reduction of Point Inflation and Fraud Risks
- Automated Multi-Channel System Announcements
- Scalable Enterprise Spring Boot Architecture

CHAPTER 3

SYSTEM REQUIREMENTS

The hardware and software requirements and the platform description of the system are detailed under sections 3.1, 3.2, and 3.3 respectively.

3.1 HARDWARE REQUIREMENTS

1. Processor Type	Intel Core i5 / AMD Ryzen 5 or higher
2. RAM	8 GB RAM minimum (16 GB recommended)
3. Hard Disk	512 GB SSD

3.2 SOFTWARE REQUIREMENTS

a. Operating System	Windows 11 / Linux / macOS
b. IDE / Development Tools	Visual Studio Code / IntelliJ IDEA
c. Front End Technologies	ReactJS (v18), Vite, CSS3, JavaScript (ES6+)
d. Back End Technologies	Java 17, Spring Boot (v3.2.5), Spring Security, Hibernate
e. Database	PostgreSQL 15+

3.3 SOFTWARE DESCRIPTION



Fig. 3.1. VS Code Logo

3.3.1 FRONTEND

ReactJS: ReactJS is a popular open-source JavaScript library for building user interfaces, particularly for single-page web applications. React follows a component-based architecture, where user interfaces are broken down into modular, reusable components. Each component encapsulates its own state and render logic, making complex UIs easier to maintain and scale. React utilizes a Virtual DOM (Document Object Model) to optimize performance by updating only changed elements in memory before modifying the actual browser DOM.

React uses JSX (JavaScript XML), allowing developers to write HTML-like structures directly within JavaScript code. The application follows a clean unidirectional data flow where state flows downwards from parent to child components via props, while user interactions trigger callbacks upward, maintaining predictable application state.

Features of ReactJS

1. **Declarative UI:** React makes it easy to create interactive UIs by automatically updating components when application state changes.
2. **Component-Based Architecture:** Reusable UI components maintain their own isolated state, enabling clean separation of concerns.
3. **Virtual DOM Optimization:** Minimizes expensive real DOM manipulation through intelligent diffing algorithms.
4. **JSX Syntax:** Combines structural HTML layout directly with JavaScript component logic.
5. **Unidirectional Data Flow:** Predictable state propagation ensures reliable data rendering across routes.

3.3.2 BACKEND Java & Spring Boot

Java 17 & Spring Boot 3.2.5: Java is an enterprise-grade, object-oriented programming language renowned for its platform independence, strong type safety, and robust memory management. Spring Boot provides an open-source Java-based framework used to build microservices and enterprise REST Web APIs. It simplifies Spring application setup by offering auto-configuration, embedded Tomcat server support, and production-ready starter dependencies.

Core Backend Extensions & Modules

1. **Spring Security Web:** Provides comprehensive authentication and role-based authorization guardrails for REST endpoints.
2. **Spring Data JPA:** Simplifies data access layer development using object-relational mapping (ORM) over PostgreSQL tables.
3. **JWT (JSON Web Token):** Enables stateless bearer token generation and verification for authenticated requests.

Key Maven Project Dependencies

1. **Spring Web:** Provides REST controller support, Jackson JSON object mapping, and HTTP request routing.
2. **Spring Boot DevTools:** Enables hot reloading and enhanced development debugging capabilities.
3. **Postgres Driver:** JDBC database driver facilitating high-performance connectivity to PostgreSQL.
4. **OpenPDF:** Open-source PDF library utilized by administrative export services to generate official audit documents.

CHAPTER 4

SYSTEM DESIGN

4.1 MODULE DESCRIPTION

- User Management & Role Governance
- Points & Tier Management Engine
- Reward & Redemption Management
- Admin Dashboard & Fraud Analytics

4.1.1 USER MANAGEMENT

The user management module governs account profiles across 4 system roles (Super Admin, Loyalty Manager, Staff, Customer). Controllers strictly enforce role hierarchy restrictions so users can only view or edit accounts within their authorized tier.

Table 4.1 User Management Schema

Field	Description
memberId	Unique primary key identifier for each member
fullName	Full name of the account holder
email	Unique email address used for login and notifications
password	Encrypted password string
role	Assigned system role (SUPER_ADMIN, LOYALTY_MANAGER, STAFF, CUSTOMER)
status	Account state (ACTIVE / INACTIVE)

4.1.2 POINTS & TIER MANAGEMENT

Calculates points allocated per purchase bill, updates total balances, calculates tiers (Bronze <1000, Silver ≥1000, Gold ≥2500, Platinum ≥5000), and tracks FIFO lot expiration.

Table 4.2 Points & Tier Calculation Schema

Field	Description
points	Current available points balance
tier	Evaluated loyalty tier (BRONZE, SILVER, GOLD, PLATINUM)
billAmount	Purchase invoice total in Indian Rupees (₹)
calculatedPoints	Points earned (floor(billAmount / 10.0))

4.1.3 REWARD & REDEMPTION MANAGEMENT

Allows customers to redeem eligible rewards based on available points, creating transaction audit records and consuming FIFO point lots.

Table 4.3 Reward & Redemption Schema

Field	Description
transactionId	Unique transaction audit record ID
transactionType	Operation type (EARN, DEDUCT, REDEEM, EXPIRY)
pointsTransacted	Number of points added or deducted
previousBalance	Points balance prior to operation
newBalance	Points balance post operation

4.1.4 ADMIN DASHBOARD & FRAUD ANALYTICS

Provides central oversight of total members, transaction ledgers, fraud flags, and admin announcements.

4.2 USE CASE DIAGRAM

A use case diagram illustrates the interactions between system actors (Customer, Staff, Loyalty Manager, Super Admin) and platform use cases.

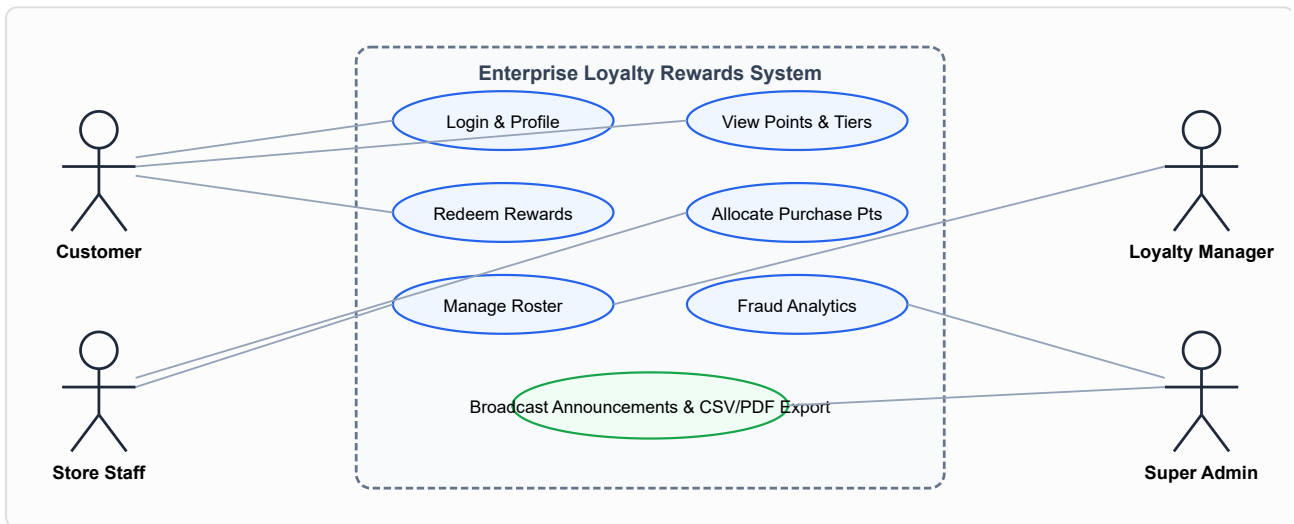


Fig. 4.1. Use Case Diagram

4.3 SEQUENCE DIAGRAM

A sequence diagram depicts the chronological message flow between React WebApp, JwtAuthenticationFilter, MemberController, MemberService, and PostgreSQL Database during authentication and point allocation requests.

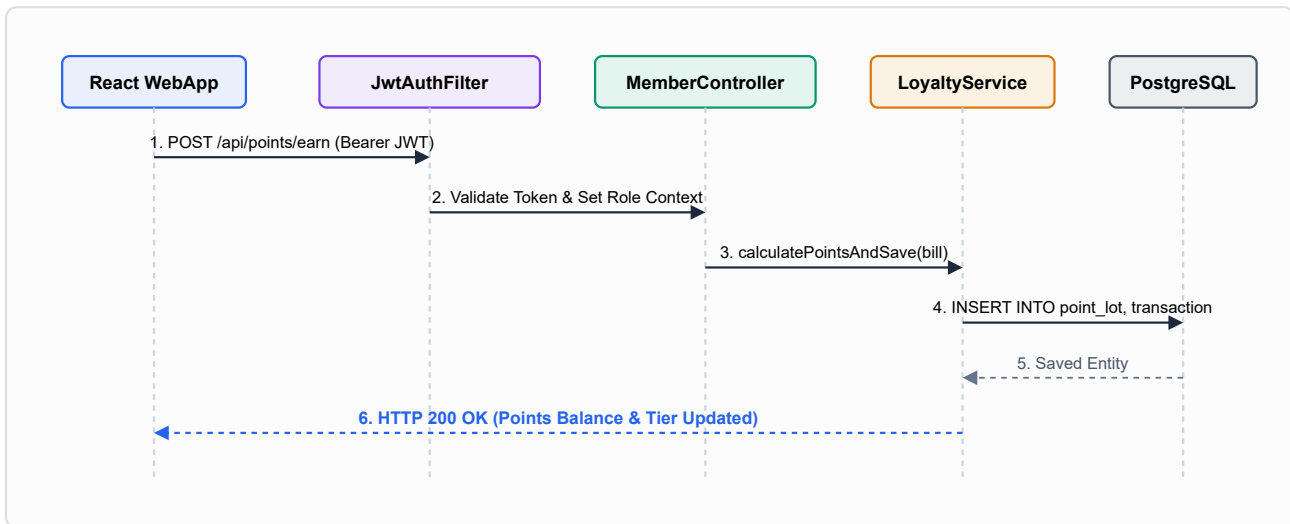


Fig. 4.2. Sequence Diagram

4.4 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) visualizes how user credentials, purchase bills, point transactions, and announcements move through system processes and data stores.

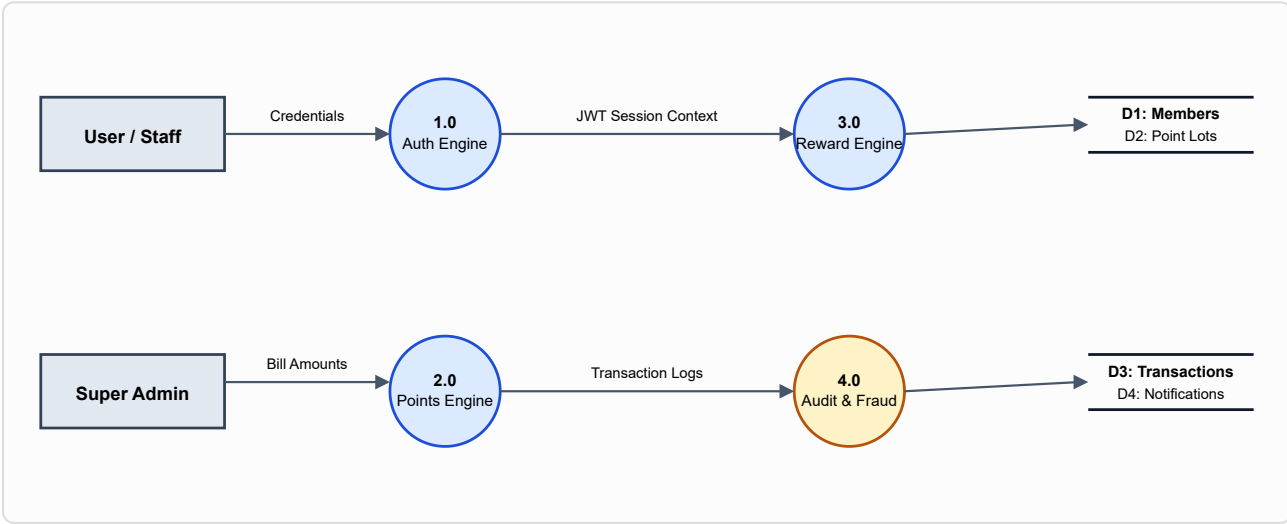


Fig. 4.3. Data Flow Diagram

CHAPTER 5

TESTING

5.1 UNIT TESTING

Unit testing forms the foundational layer of the testing process for the Enterprise Loyalty Rewards Management System project. It involves testing individual Java service methods, repository queries, and utility functions in isolation to identify issues early. Unit tests validate point calculations, FIFO lot deductions, and password hashing logic.

5.2 INTEGRATION TESTING

Integration testing validates seamless interaction between REST API endpoints, Spring Security filters, and PostgreSQL database queries. API endpoint responses for all 4 roles are tested to verify proper JSON response structure and HTTP status codes.

5.3 SECURITY AND AUTHENTICATION

JSON Web Tokens (JWTs) provide stateless user authentication. Upon successful login at `/api/members/login`, a signed JWT is issued containing user email, member ID, full name, and assigned role.

Requests include this bearer token in the Authorization header. The Spring Security filter chain validates token integrity and assigns role authorities (ROLE_SUPER_ADMIN, ROLE_LOYALTY_MANAGER, ROLE_STAFF, ROLE_CUSTOMER).

```
// Store JWT Token & Role Context in LocalStorage upon login
const response = await axios.post('/api/members/login', { email, password });
if (response.data.token) {
  localStorage.setItem('token', response.data.token);
  localStorage.setItem('role', response.data.role);
  localStorage.setItem('memberName', response.data.fullName);
  navigate('/dashboard');
}
```

Fig. 5.1. Storing the Token in Local Storage


```
// Axios Interceptor attaching Bearer token to API Requests
axios.interceptors.request.use(config => {
  const token = localStorage.getItem('token');
  if (token) {
    config.headers.Authorization = `Bearer ${token}`;
  }
  return config;
});
```

Fig. 5.2. Authenticating the User using Bearer Token

5.4 TEST CASES

Test cases evaluate system functionality including authentication, role enforcement, point allocation, and announcement dispatch.

5.4.1 TEST CASE I: Role-Based Authorization Enforcement

403 Forbidden - Access Denied

User role ROLE_LOYALTY_MANAGER is not authorized to access endpoint GET /api/admin/audit-logs.

Fig. 5.3. Test Case I

EXPECTED OUTPUT: System blocks unauthorized access and returns HTTP 403 Forbidden.

ACTUAL OUTPUT: Backend rejects request with 403 Forbidden: Access Denied.

5.4.2 TEST CASE II: Admin Announcement Dispatch

System Announcement Broadcast Successful

Announcement titled *"Quarterly Bonus Points Campaign"* sent to 1 Loyalty Managers and 9 Staff Members.

Fig. 5.4. Test Case II

EXPECTED OUTPUT: Announcement is created for all active managers and staff, excluding customers.

ACTUAL OUTPUT: System displays ✓ Announcement sent to 1 managers and 9 staff members. Announcement appears in manager/staff notification bell dropdown with 🔔 icon and is hidden for customers.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The Enterprise Loyalty Rewards Management System project successfully addresses the challenges of customer retention, reward accounting, and security governance in retail commerce. By providing a full-stack platform built with ReactJS, Java Spring Boot, and PostgreSQL, the system delivers mathematical transparency, real-time transaction auditability, and automated point expiry management. The implementation of strict backend Role-Based Access Control (RBAC) ensures that privacy and authorization boundaries are enforced across all 4 system roles, eliminating unauthorized account access and fraud exposure. Overall, this project represents a significant step forward in modern digital customer loyalty solutions.

6.2 FUTURE WORK

Future enhancements could include developing native iOS and Android mobile applications using React Native to offer mobile push notifications and barcode scanning for instant store checkout point crediting. Additionally, integrating machine learning models for predictive customer churn analysis and personalized reward recommendations would further enhance commercial value for retail enterprises.

CHAPTER 7

APPENDICES

APPENDIX I: SOURCE CODE

MemberController.java (Excerpt)

```
package com.loyalty.rewards.controller;

import com.loyalty.rewards.model.Member;
import com.loyalty.rewards.model.Role;
import com.loyalty.rewards.service.MemberService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.*;
import org.springframework.web.server.ResponseStatusException;

@RestController
@RequestMapping("/api/members")
public class MemberController {

    @Autowired
    private MemberService memberService;

    @GetMapping("/{id}")
    public ResponseEntity<Member> getMemberById(@PathVariable Long id, Authentication authentication) {
        Member current = (Member) authentication.getPrincipal();
        if (current.getRole() == Role.CUSTOMER && !current.getMemberId().equals(id)) {
            throw new ResponseStatusException(HttpStatus.FORBIDDEN, "Customers can only view own profile.");
        }
        Member target = memberService.getMemberById(id);
        if (current.getRole() == Role.STAFF && target.getRole() != Role.CUSTOMER) {
            throw new ResponseStatusException(HttpStatus.FORBIDDEN, "STAFF can view CUSTOMER profiles only.");
        }
        if (current.getRole() == Role.LOYALTY_MANAGER && target.getRole() == Role.SUPER_ADMIN) {
            throw new ResponseStatusException(HttpStatus.FORBIDDEN, "LOYALTY_MANAGER cannot view SUPER_ADMIN details.");
        }
        return ResponseEntity.ok(target);
    }
}
```

```
}  
}
```

NotificationController.java (Announcement Excerpt)

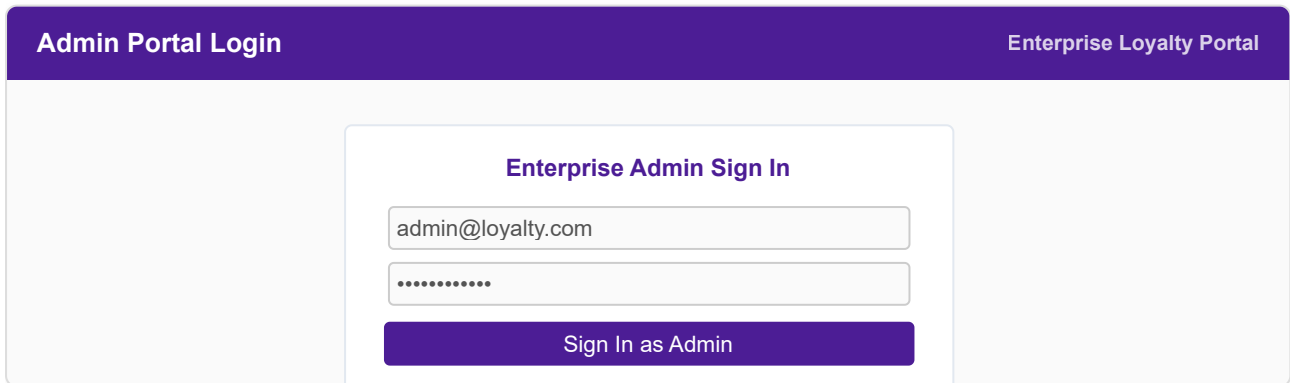
```
@PostMapping("/announcement")
public ResponseEntity<Map<String, Object>> sendAnnouncement(
    @RequestBody Map<String, String> body,
    Authentication authentication) {

    Member current = (Member) authentication.getPrincipal();
    if (current == null || current.getRole() != Role.SUPER_ADMIN) {
        throw new ResponseStatusException(HttpStatus.FORBIDDEN, "Access Denied: Only SUPER_ADMIN users can send system announcements.");
    }

    String title = body.get("title");
    String message = body.get("message");

    Map<String, Object> result = notificationService.sendAnnouncement(title,
message, current);
    return ResponseEntity.ok(result);
}
```


APPENDIX II: SCREENSHOTS



Admin Portal Login

Enterprise Loyalty Portal

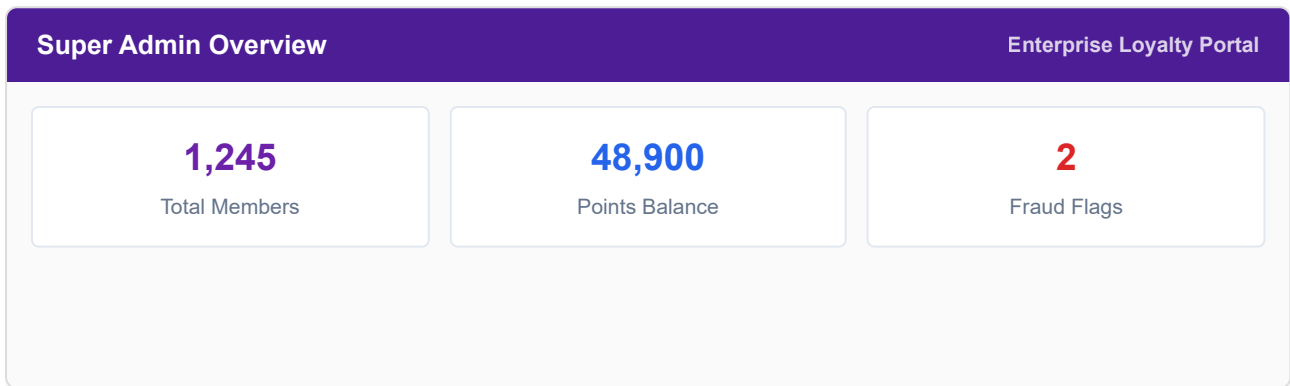
Enterprise Admin Sign In

admin@loyalty.com

.....

Sign In as Admin

Fig. A.2.1. Admin Login



Super Admin Overview

Enterprise Loyalty Portal

1,245
Total Members

48,900
Points Balance

2
Fraud Flags

Fig. A.2.2. Admin Dashboard

Member Roster & Governance				Enterprise Loyalty Portal
ID	Name	Email	Role	Tier
#101	Karthick G	karthick@example.com	SUPER_ADMIN	Platinum
#102	Gayathri K	gayathri@example.com	STAFF	Gold

Fig. A.2.3. Member Management Roster

Points Allocation Portal		Enterprise Loyalty Portal
Select Member: Karthick Geethanath (#101) Purchase Bill Amount (₹): ₹ 2,500.00		Calculated Points Earned +250 PTS Confirm & Credit Points

Fig. A.2.4. Earn & Deduct Points Portal

Transaction Audit Ledger				Enterprise Loyalty Portal	
Audit Records (Filter: EARN / DEDUCT)				Export CSV	Export PDF Report
Tx ID	Type	Bill ₹	Points	Performed By	Timestamp
#TX-8801	EARN	₹ 2,500	+250	Store Staff #102	2025-08-15 14:30
#TX-8802	REDEEM	-	-100	Customer #101	2025-08-15 16:10

Fig. A.2.5. Transaction Ledger Audit & Export

Customer Login Page	Enterprise Loyalty Portal
Customer Loyalty Portal customer@example.com ***** Sign In	

Fig. A.2.6. User Login Page

Customer Registration Page

Enterprise Loyalty Portal

Join Enterprise Rewards

Karthick Geethanath

karthick@example.com

Register Account

Fig. A.2.7. User Registration Page

Customer Rewards Dashboard

Enterprise Loyalty Portal

Available Points

1,450 PTS

Tier: Platinum 🏆

Next Tier: Diamond (550 pts away)

Fig. A.2.8. User Dashboard

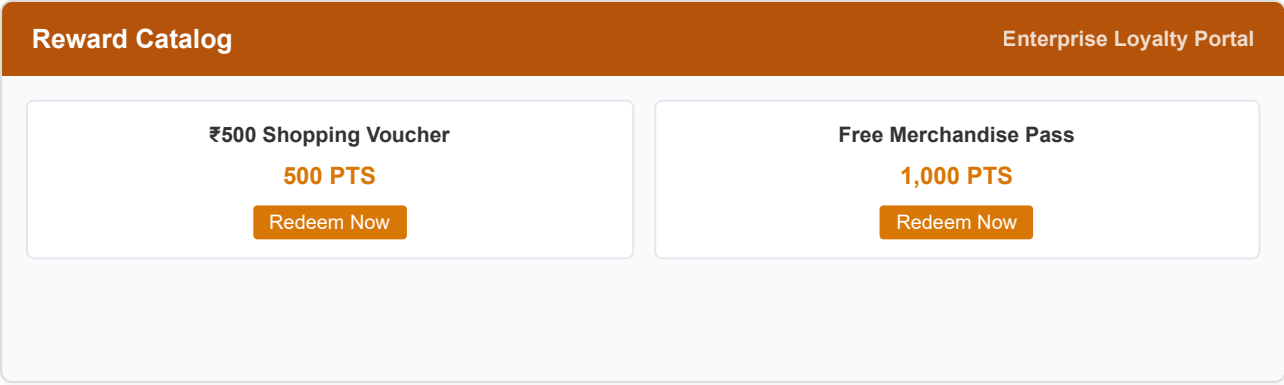


Fig. A.2.9. Reward Catalog & Redemption

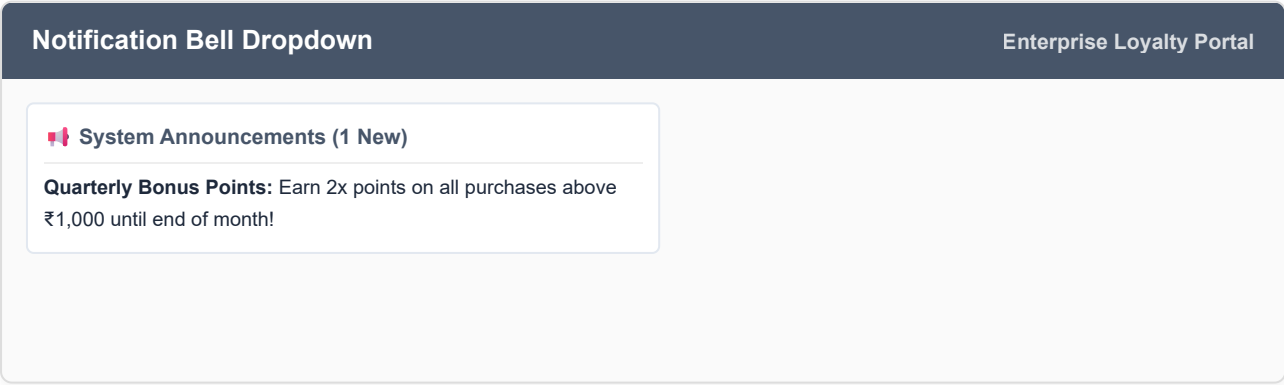


Fig. A.2.10. System Announcement Bell Dropdown

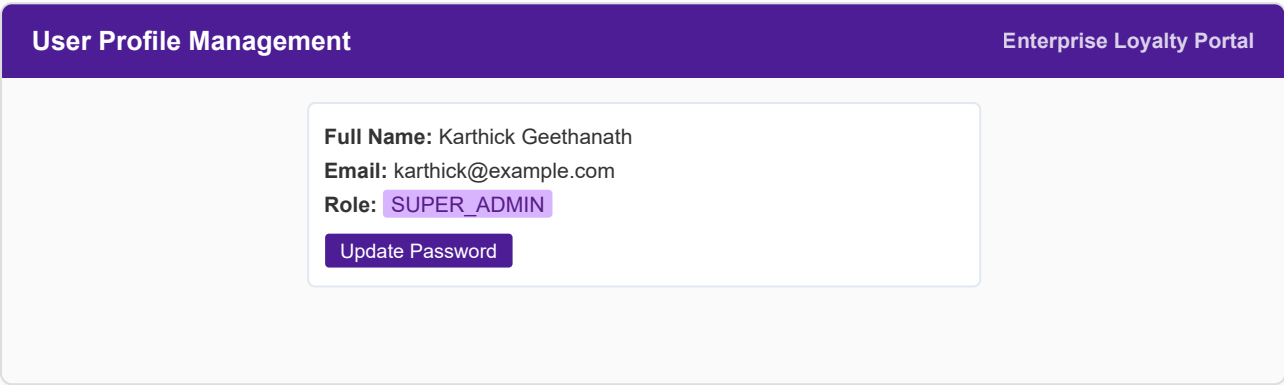


Fig. A.2.11. User Profile Management

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